

Express Setup Cheatsheet

Packages, project structure, Swagger & Postman quick reference

1. Packages Installed

Package	Purpose
express	Core web framework — routing, middleware, request/response handling
nodemon (dev)	Auto-restarts server on file changes
dotenv	Loads variables from .env file into process.env
morgan	Logs incoming requests (method, URL, status, time) in dev mode
superagent	HTTP client for making requests to external APIs
swagger-ui-express	Renders Swagger/OpenAPI docs as an interactive page at /api-docs
swagger-autogen	Scans routes and auto-generates swagger-output.json
yamljs	(Optional) Loads a manual swagger.yaml file, if not using autogen

```
npm install express morgan superagent dotenv swagger-ui-express swagger-autogen
npm install -D nodemon
```

2. Project Structure

```
ExpressJs/ .env(PORT=3000) server.js(app.listen) app.js(middleware+routes)
  nodemon.json({"ext":"js,json,yaml"}) swagger-generate.js swagger-output.json
```

3. app.js — Core Pattern

```
const express = require('express');
const morgan = require('morgan');
const swaggerUi = require('swagger-ui-express');
const swaggerFile = require('./swagger-output.json');
const app = express();

if (process.env.NODE_ENV === 'development') app.use(morgan('dev'));
app.use(express.json());
app.use('/api-docs', swaggerUi.serve, swaggerUi.setup(swaggerFile));

app.get('/api/v1/users', (req, res) => {
  res.status(200).json({ status: 'success', message: 'All users fetched' });
});
// POST -> 201 Created | PATCH -> 200 OK | DELETE -> 204 No Content

app.use((req, res) => { // 404 handler — must be LAST
  res.status(404).json({ status: 'fail', message: `Route ${req.originalUrl} not found` });
});
module.exports = app;
```

4. Swagger — Auto-Generate Workflow

```
// swagger-generate.js
const swaggerAutogen = require('swagger-autogen')();
swaggerAutogen('./swagger-output.json', ['./app.js'],
  { info: { title: 'My Learning API' }, host: 'localhost:3000' });

// Run whenever routes change: node swagger-generate.js (or npm run swagger)
// Then refresh: http://localhost:3000/api-docs
// Optional comments per route: #swagger.summary / #swagger.tags
```

5. Postman — Skills to Build Alongside the Course

Environments ({{base_url}}) → Collections/Folders → Pre-request Scripts (auth tokens) → Tests tab (auto-check status codes) → Save response data as variables (e.g. login token reused in later requests).

6. When to Use What (Real-World)

Project size	Docs approach
Small/personal	Postman collection only
Medium (JS)	swagger-autogen + comments
Large (TypeScript)	tsoa / zod-to-openapi
Enterprise	Schema-first OpenAPI contract